

EXPLAINABLE ARTIFICIAL INTELLIGENCE

L193 – Lecture 6 – Lent 2025



UNIVERSITY OF
CAMBRIDGE



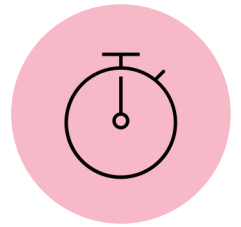
NEURAL INTERPRETABLE REASONING

Pietro Barbiero, PhD

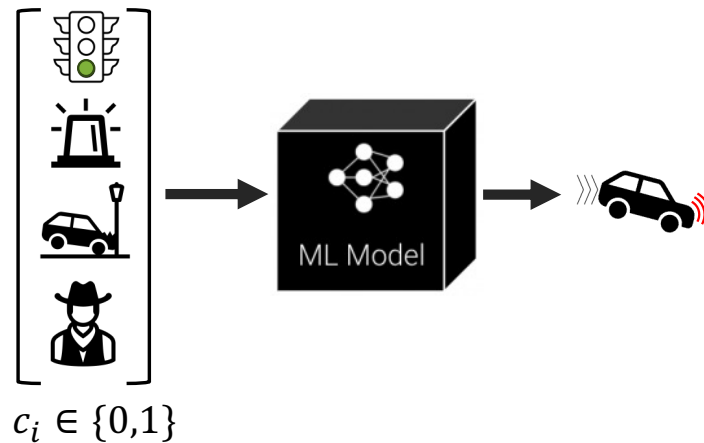
IBM Research



THE COMPLEXITY OF UNDERSTANDING



Are <20 interactions sufficient to fully understand this ML model?

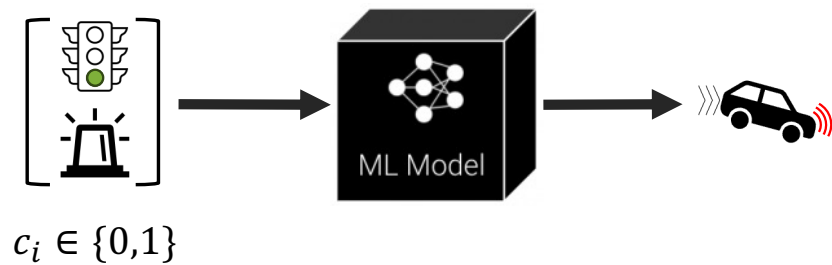


If yes, raise
your hand!

Disclaimer: Unfortunately for you, I'm always right during the lecture—but I'm happy to be proven wrong afterward! 😊

THE COMPLEXITY OF UNDERSTANDING

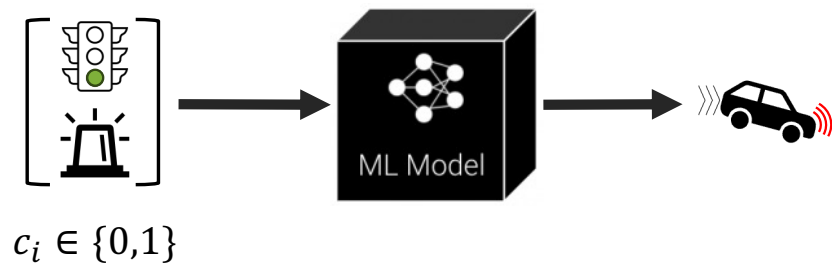
How many interactions do we need to fully understand the model?



0	0	1

THE COMPLEXITY OF UNDERSTANDING

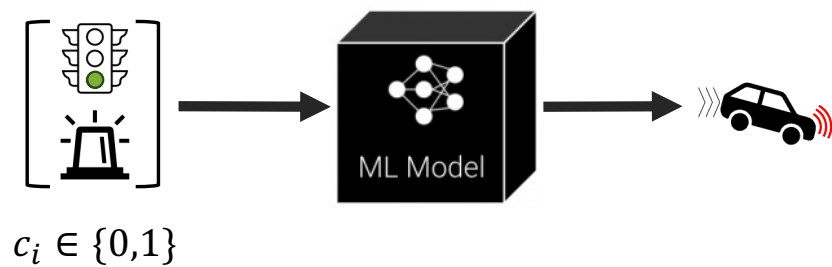
How many interactions do we need to fully understand the model?



			
0	0	1	green light=1 ↩
1	0	0	

THE COMPLEXITY OF UNDERSTANDING

How many interactions do we need to fully understand the model?



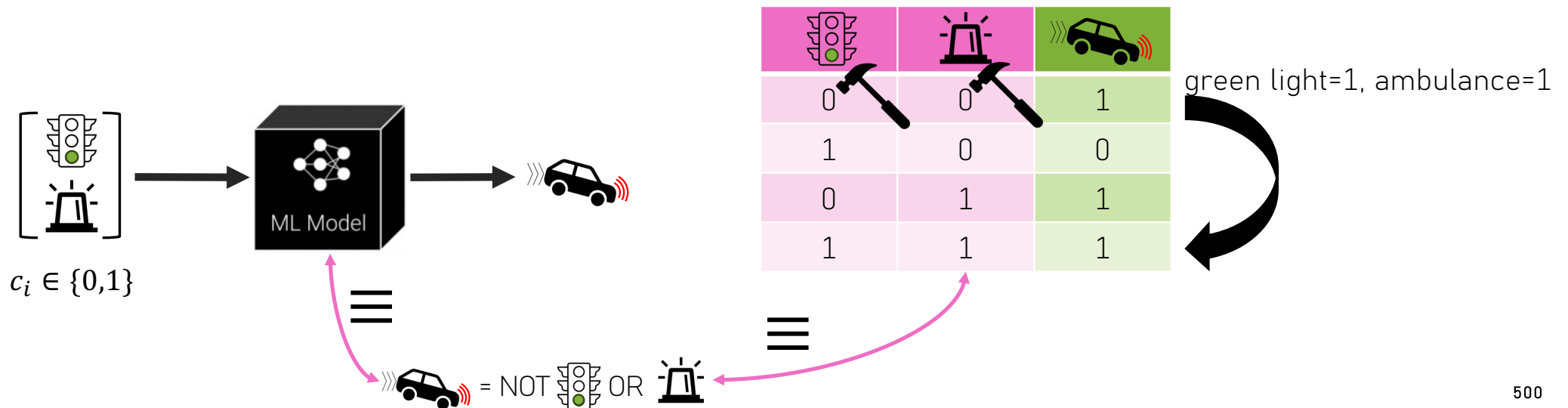
0	0	1
1	0	0
0	1	1

ambulance=1

THE COMPLEXITY OF UNDERSTANDING

How many interactions do we need to fully understand the model?

We can **recover the full Conditional Probability / Truth Table!**



THE COMPLEXITY OF UNDERSTANDING

How many interactions do we need to fully understand the model?

We can recover the full Conditional Probability / Truth Table!

*#interactions required to extract full CPT/TT
is exponential in #inputs!*

Can we do better?



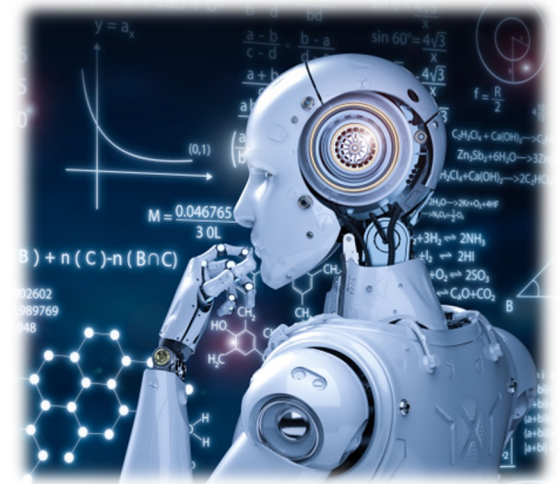
(INFORMAL) OBJECTIVE

Build a **general-purpose** neural model that is:

❑ **Expressive** (as DNNs)

and whose inference mechanism is

- ❑ **Functionally transparent** (we fully understand CPT/TT)
- ❑ **Tractable** (CPT/TT size $\ll$ exp)
- ❑ **Semantically transparent** (concept-based)
- ❑ **Causally transparent** (based on non-trivial cause-effect chains)



PLAN

- General methods (foundations)
- Neural Interpretable Reasoning paradigm
- Grounding NIR (Concept Memory Reasoning example)
- Causal reasoning (metrics and models)



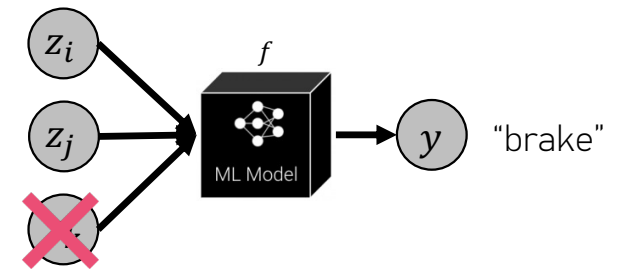
METHODS

Functional transparency: understanding of inference mechanism is tractable

$$|\text{CPT}(f)| \ll \exp(|Z|)$$

Methods:

- Filter irrelevant features out



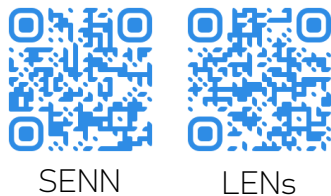
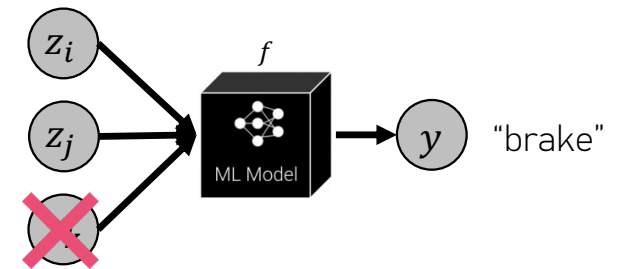
METHODS

Functional transparency: **understanding** of inference mechanism is **tractable**

$$|\text{CPT}(f)| \ll \exp(|Z|)$$

Methods:

- Filter **irrelevant** features out
- **Re-parametrize** global CPT as a **mixture** of simple (e.g., linear) “local CPTs”



$$\text{Global CPT} \quad \text{NOT } z_i \text{ OR } z_j = \text{OR} \left\{ \begin{array}{c} \text{NOT } z_i \\ z_j \end{array} \right\}$$

Local CPT



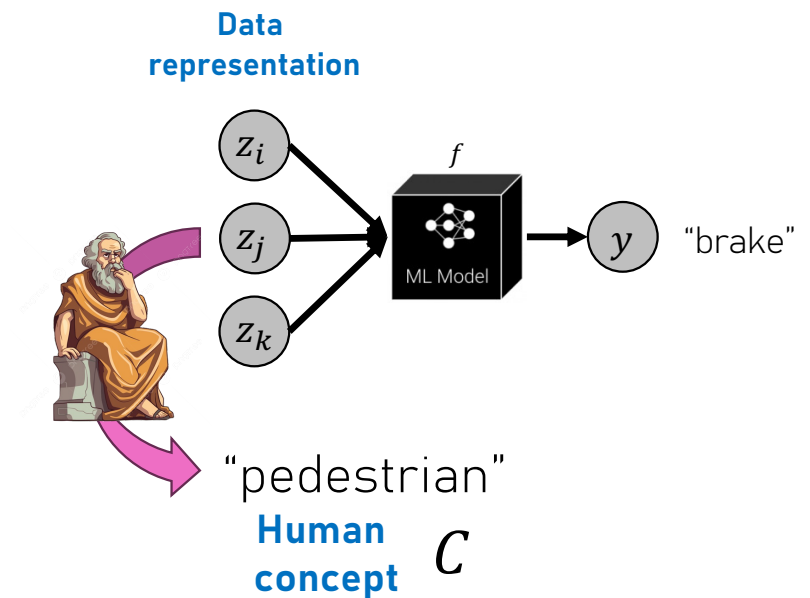
METHODS

Semantic transparency: input features are **aligned** with human semantics

Methods:

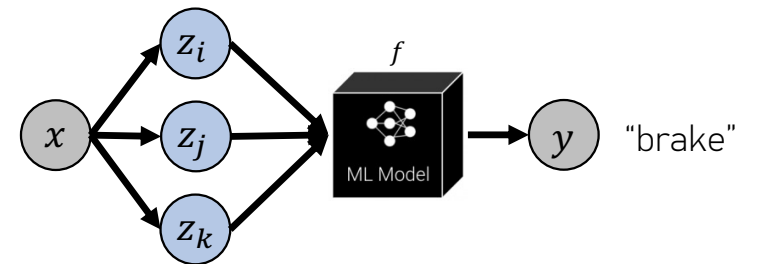
- Work with **tabular** data
- Use **concept-based** approach

... and then write the model as $P(Y | C)$!



METHODS

General-purpose: model design should be applicable to any data type



METHODS

General-purpose: model design should be applicable to any data type

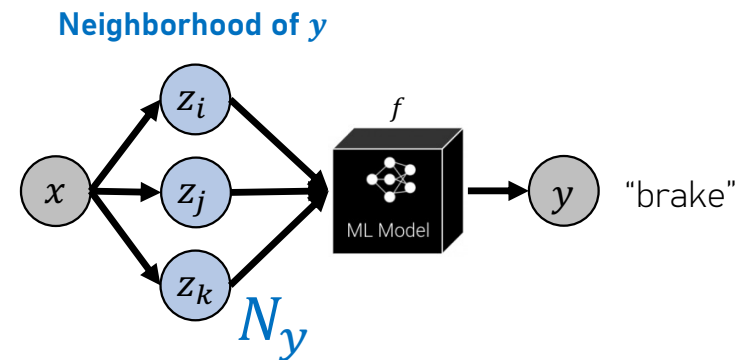
Interpretability is a **Markovian property**:

$$(Y \perp X) \mid N_y$$

Methods:

- **Re-parametrize** the inputs of an ML model f **without affecting its interpretability** (idea behind CBMs)!

$$P(Y \mid X) = \sum_c P(Y \mid C)P(C \mid X)$$

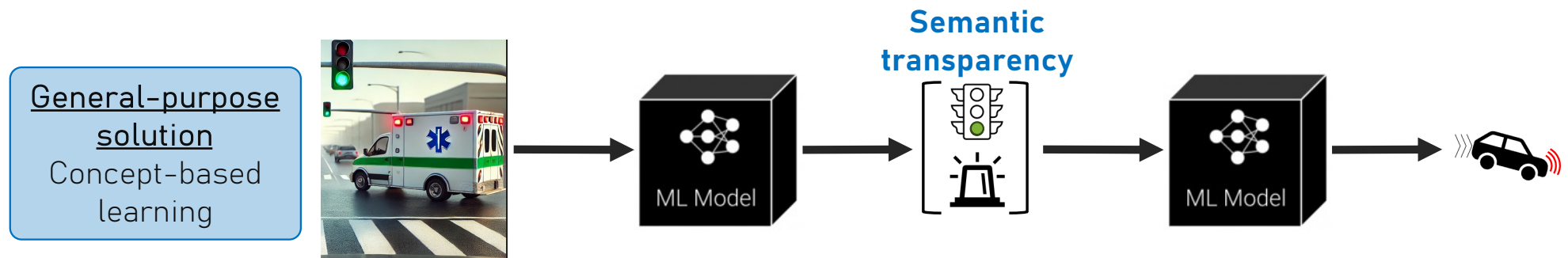


Introducing x **does not affect** the understanding of f !



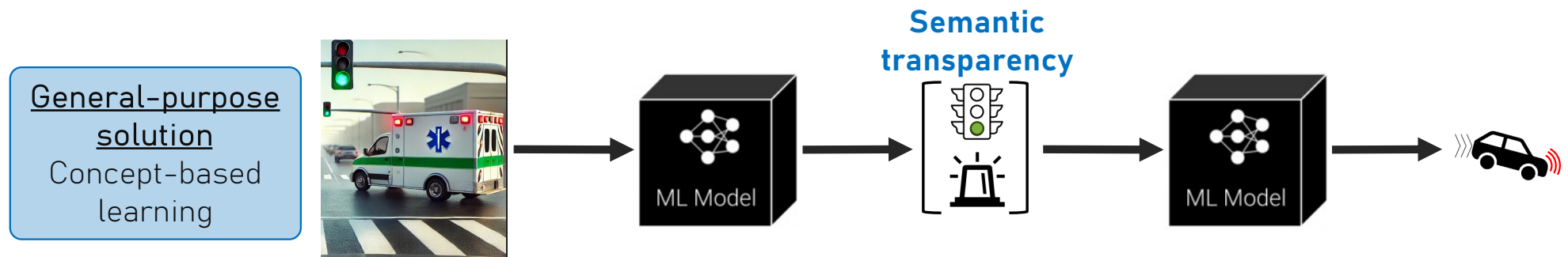
SEMANTIC & FUNCTIONAL OPACITY

Semantic opacity: data representations are not aligned with human concepts

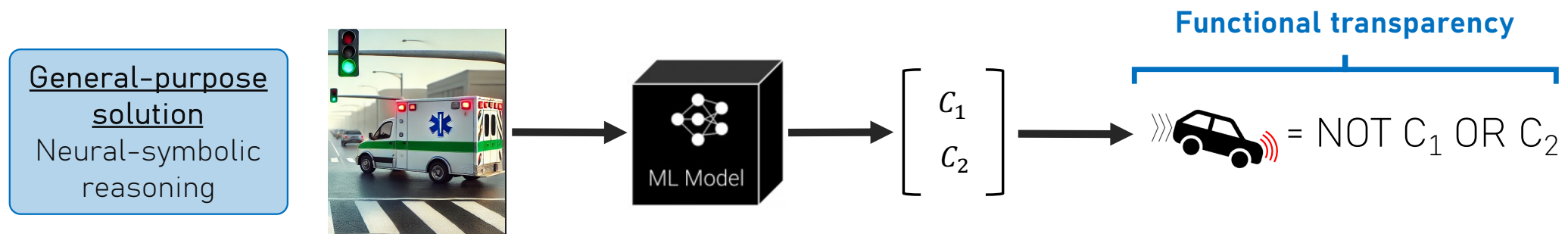


SEMANTIC & FUNCTIONAL OPACITY

Semantic opacity: data representations are not aligned with human concepts



Functional opacity: CPT is unknown or intractable to reconstruct



SEMANTIC & FUNCTIONAL OPACITY

Does **semantic** transparency imply **functional** transparency?

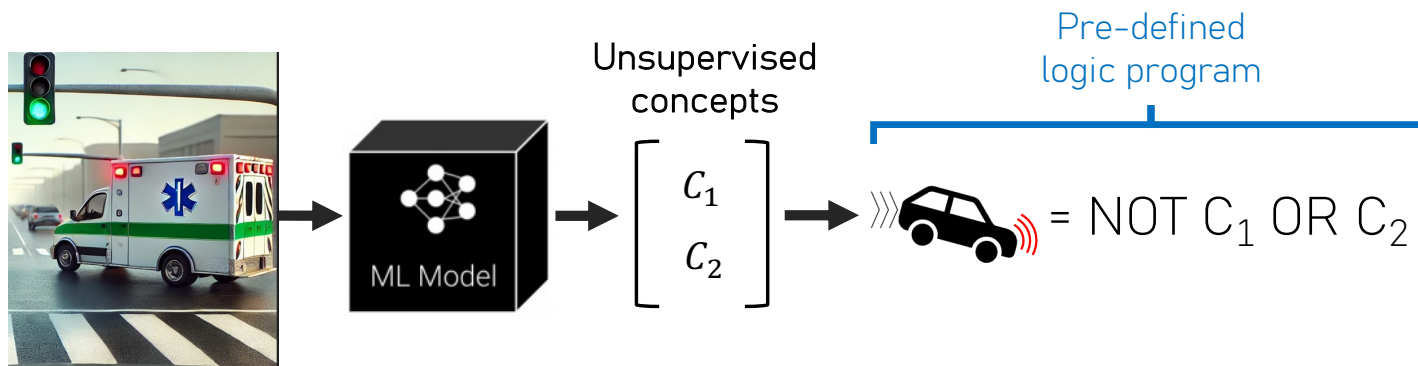
In theory it does, but...

... with n inputs the size of a TT is 2^n !
→ **intractable!**

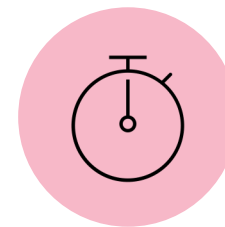
SEMANTIC & FUNCTIONAL OPACITY

Does **functional** transparency imply **semantic** transparency?

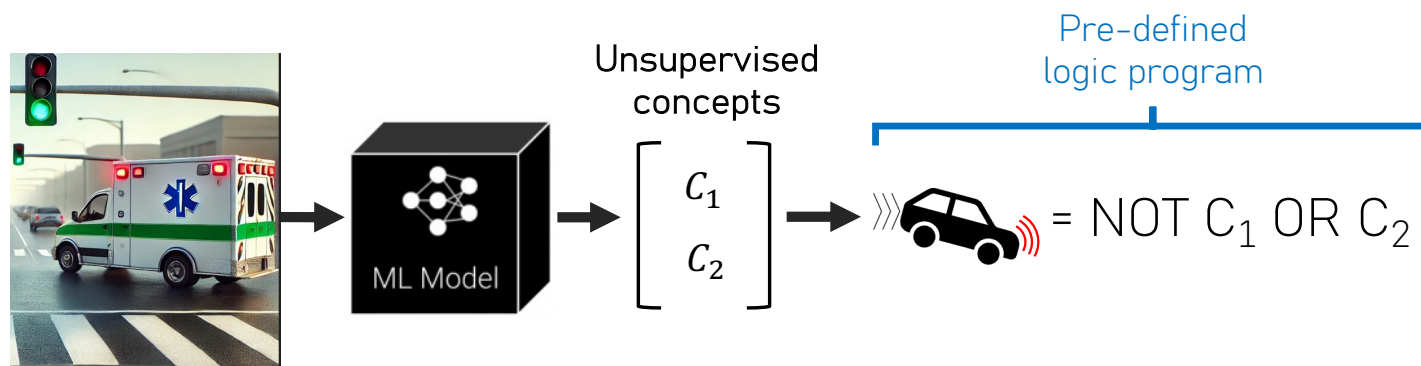
Potential solution: DeepProbLog



FUNCTIONAL $\Rightarrow$ SEMANTIC TRANSP.?



Can we prove that the classification head is safe?

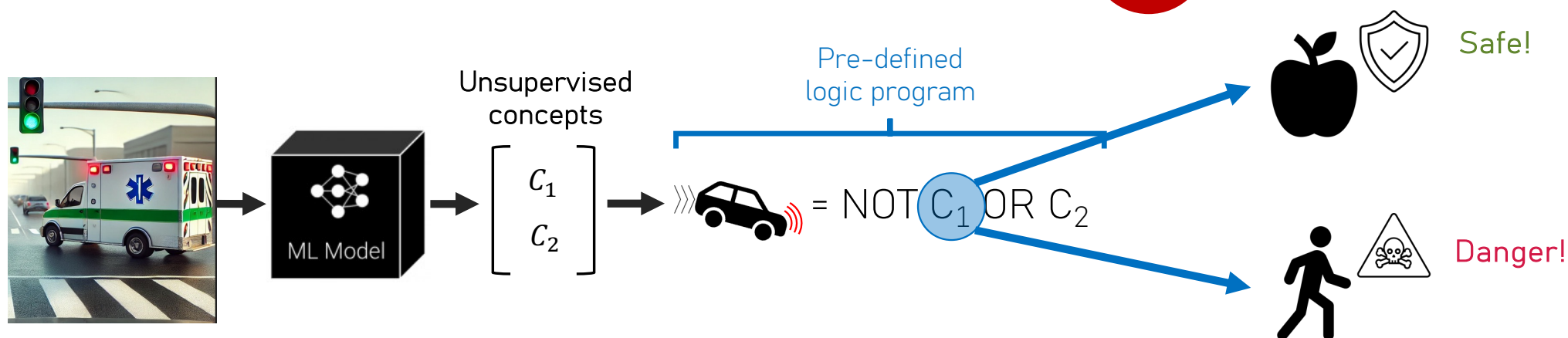


If yes, raise your hand!

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FUNCTIONAL $\Rightarrow$ SEMANTIC TRANSP.?

Can we prove that the classification head is safe?



Safety depends on concepts' meaning!

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FUNCTIONAL $\Rightarrow$ SEMANTIC TRANSP.?

Can we prove that the classification head is safe?



Unsupervised
concepts

Pre-defined
logic program



Safe!

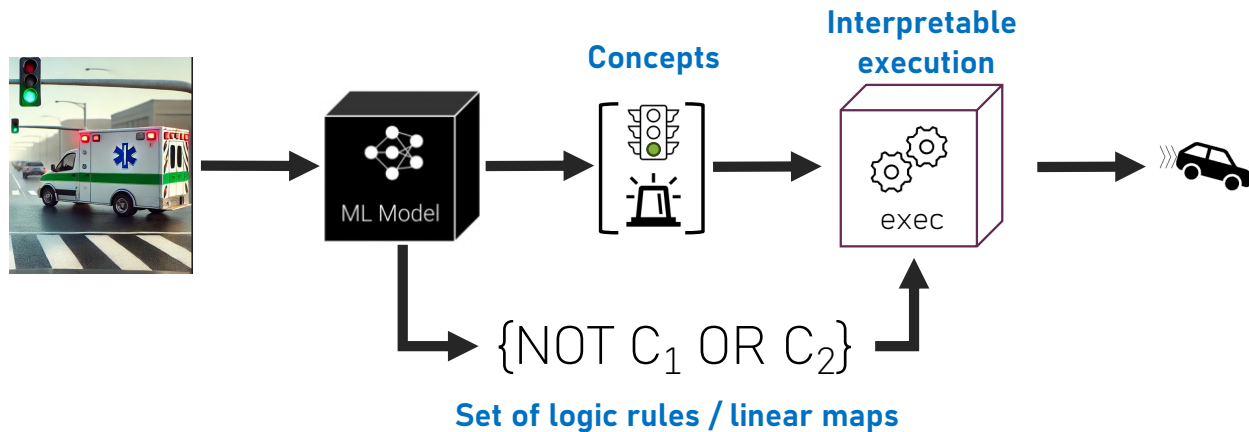
*Can we combine concept-based learning
with symbolic reasoning?*

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NEURAL INTERPRETABLE REASONING

Proposed Solution

- Step 1:** DNN generates both concept activations & rule parameters (*neural generation*)
- Step 2:** Symbolic engine executes the rule using concept activations (*interpretable execution*)



ICML23



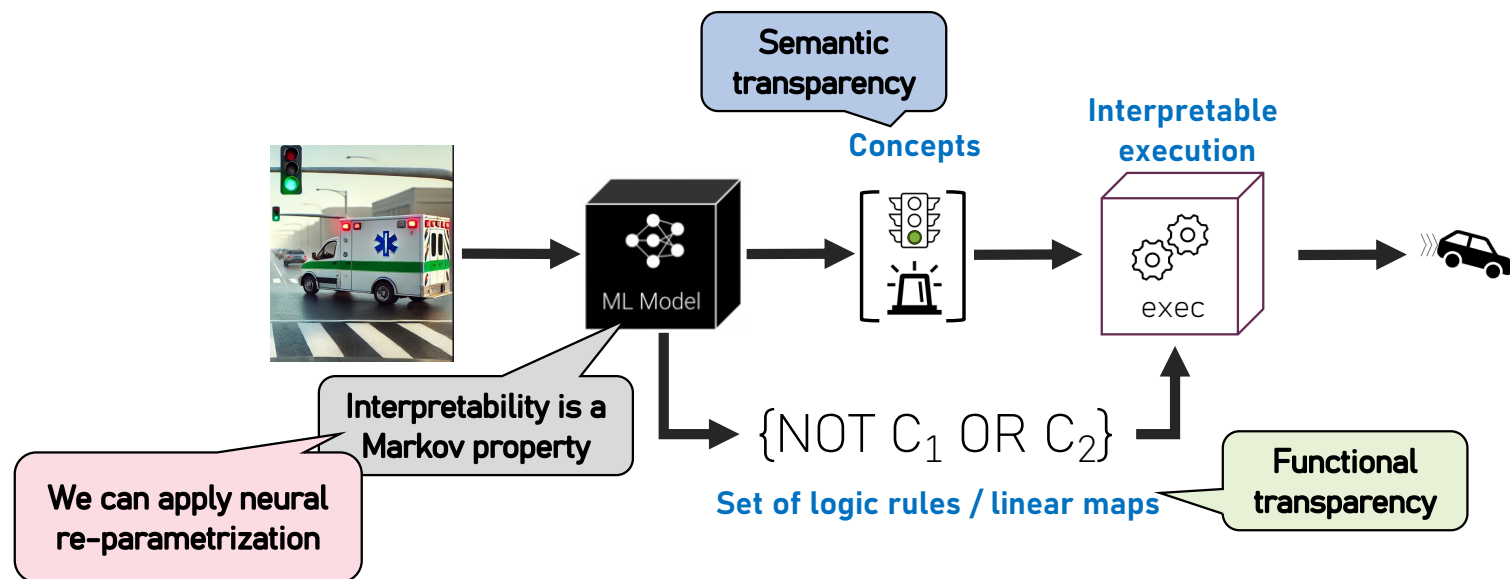
NeurIPS24

- [1] Barbiero, Pietro, et al. "Interpretable neural-symbolic concept reasoning." International Conference on Machine Learning. PMLR, 2023.
- [2] Debot, David, et al. "Interpretable concept-based memory reasoning." NeurIPS 2024.

NEURAL INTERPRETABLE REASONING

Proposed Solution

$$P(Y | X) = P(Y | \textcolor{blue}{C}; \textcolor{green}{v} \{ \textcolor{green}{W}_i \}) \textcolor{red}{P}(\textcolor{red}{C}, \{ \textcolor{red}{W}_i \} | X)$$



ICML23



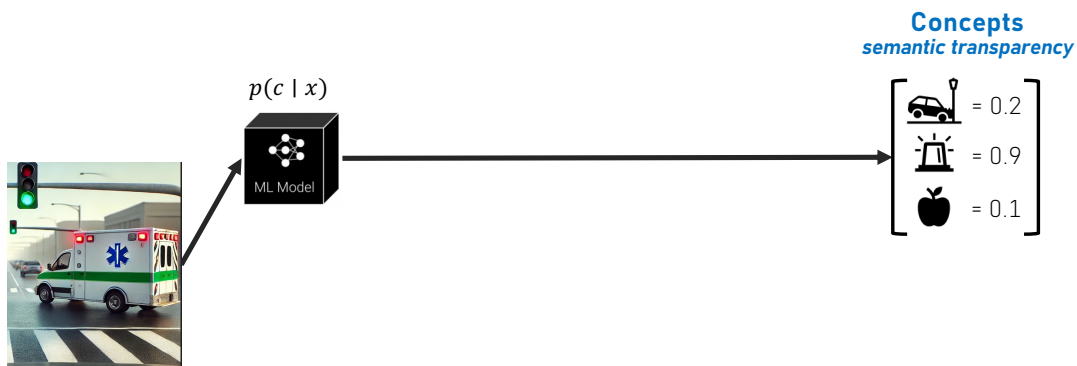
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CONCEPT-BASED MEMORY REASONING

Proposed Solution

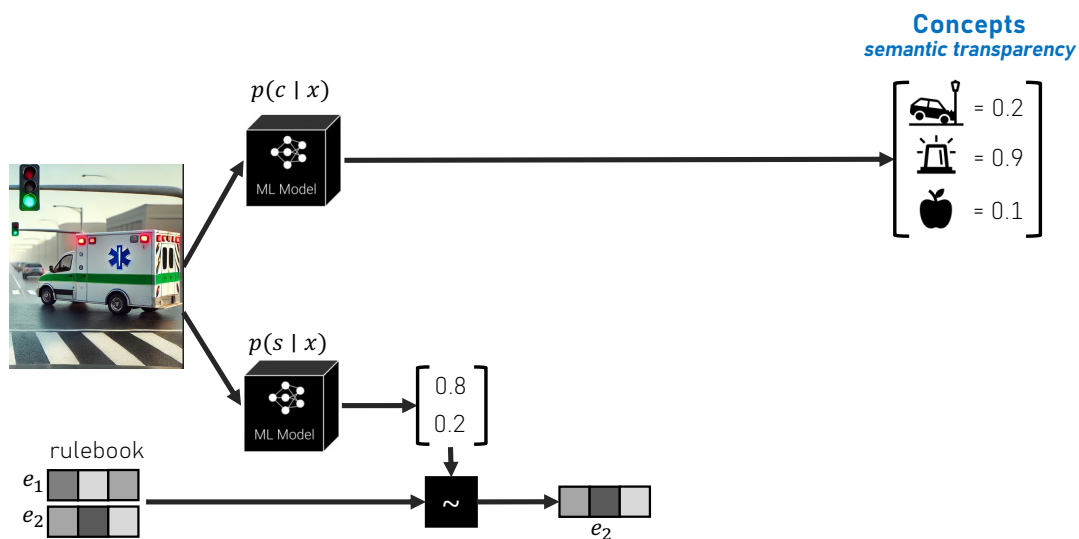
Step 1: DNN predicts concept activations



CONCEPT-BASED MEMORY REASONING

Proposed Solution

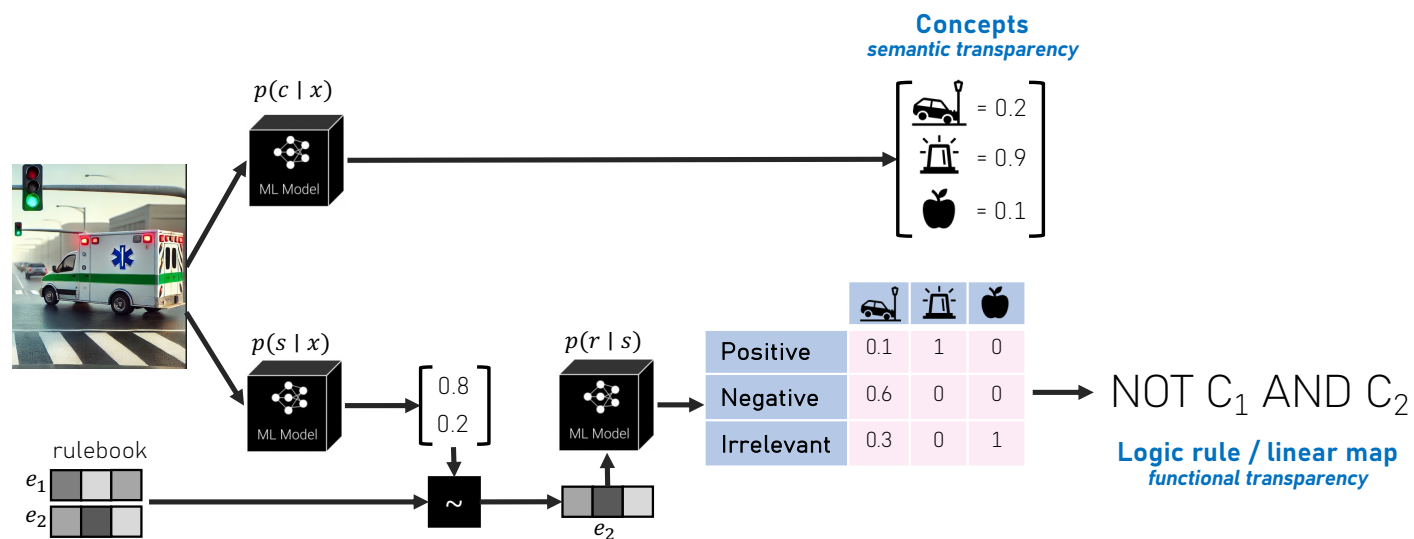
Step 2: DNN predicts embedding to be selected from the latent rulebook



CONCEPT-BASED MEMORY REASONING

Proposed Solution

Step 3: DNN decodes selected embedding into 3 states: positive, negative, irrelevant



CONCEPT-BASED MEMORY REASONING

Proposed Solution (details)

Step 4: Execute rule using rule states and concept activations

Concepts

$$\begin{bmatrix} \text{car} = 0.2 \\ \text{bell} = 0.9 \\ \text{apple} = 0.1 \end{bmatrix}$$

Rule states

NOT C_1 AND C_2



CONCEPT-BASED MEMORY REASONING

Proposed Solution (details)

Step 4: Execute rule using rule states and concept activations

Execution of rule states on concept activations

Concepts

$$\begin{bmatrix} \text{car} = 0.2 \\ \text{bell} = 0.9 \\ \text{apple} = 0.1 \end{bmatrix}$$

Rule states

NOT C_1 AND C_2

$$\mathbb{I}[r_i = \text{positive}]p(c_i | x) + \mathbb{I}[r_i = \text{negative}]p(\neg c_i | x) + \mathbb{I}[r_i = \text{irrelevant}]$$



CONCEPT-BASED MEMORY REASONING

Proposed Solution (details)

Step 4: Execute rule using rule states and concept activations

Execution of rule states on concept activations

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 $0 \times p(c_i | x) + 1 \times p(\neg c_i | x) + 0 = p(\neg c_i | x) = (1 - 0.2) = 0.8$



CONCEPT-BASED MEMORY REASONING

Proposed Solution (details)

Step 4: Execute rule using rule states and concept activations

Execution of rule states on concept activations

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$$0 \times p(c_i | x) + 1 \times p(\neg c_i | x) + 0 = p(\neg c_i | x) = (1 - 0.2) = 0.8$$

$$1 \times p(c_i | x) + 0 \times p(\neg c_i | x) + 0 = p(c_i | x) = 0.9$$

Rule states

NOT C_1 AND C_2



CONCEPT-BASED MEMORY REASONING

Proposed Solution (details)

Step 4: Execute rule using rule states and concept activations

Execution of rule states on concept activations

Concepts

$$\begin{bmatrix} \text{car} = 0.2 \\ \text{bell} = 0.9 \\ \text{apple} = 0.1 \end{bmatrix}$$

Rule states

NOT C_1 AND C_2

$$\mathbb{I}[r_i = \text{positive}]p(c_i | x) + \mathbb{I}[r_i = \text{negative}]p(\neg c_i | x) + \mathbb{I}[r_i = \text{irrelevant}]$$

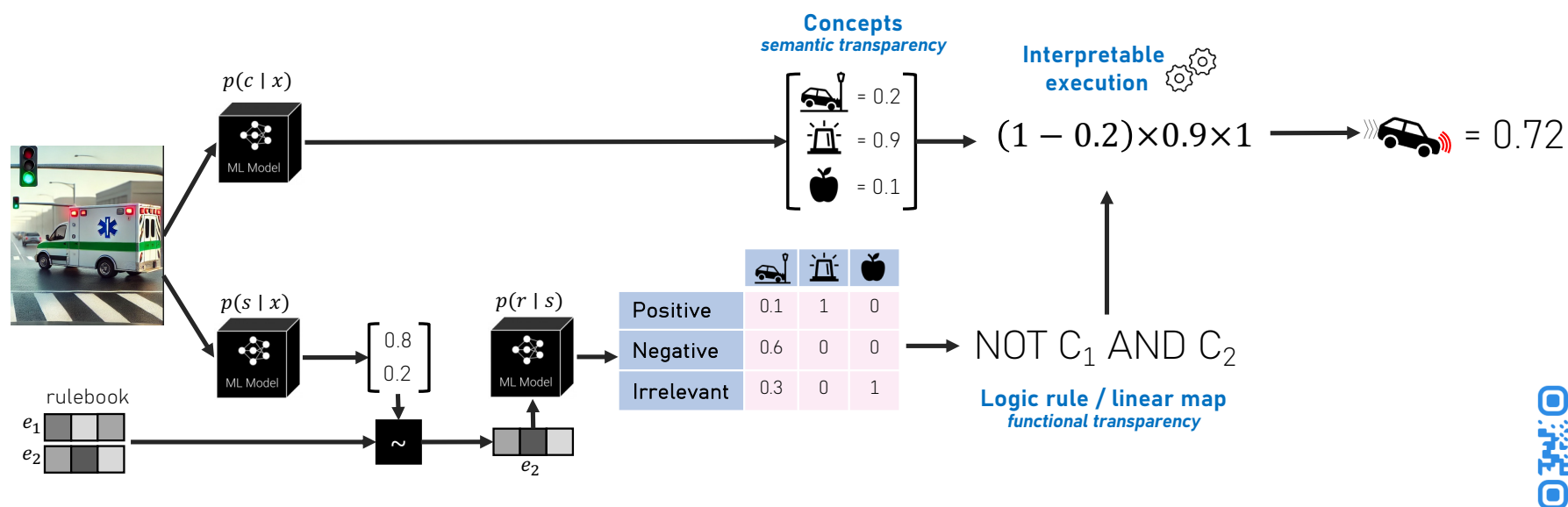
	$0 \times p(c_i x) + 1 \times p(\neg c_i x) + 0 = p(\neg c_i x) = (1 - 0.2) = 0.8$
	$1 \times p(c_i x) + 0 \times p(\neg c_i x) + 0 = p(c_i x) = 0.9$
	$0 \times p(c_i x) + 0 \times p(\neg c_i x) + 1 = 1$



CONCEPT-BASED MEMORY REASONING

Proposed Solution

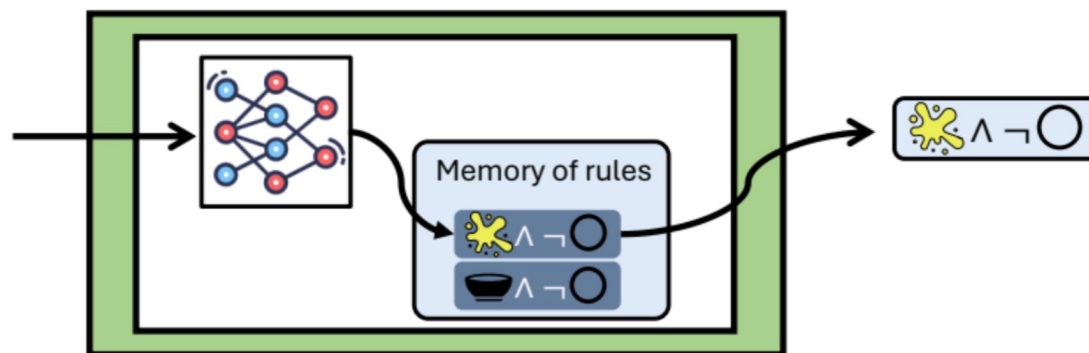
Step 5: Execute the rule combining concept states and activations to predict the output label



CONCEPT-BASED MEMORY REASONING

CMR has 3 key features:

- **Universal approximator** akin to opaque DNNs (Theorem 4.1)



Mixture of linear models

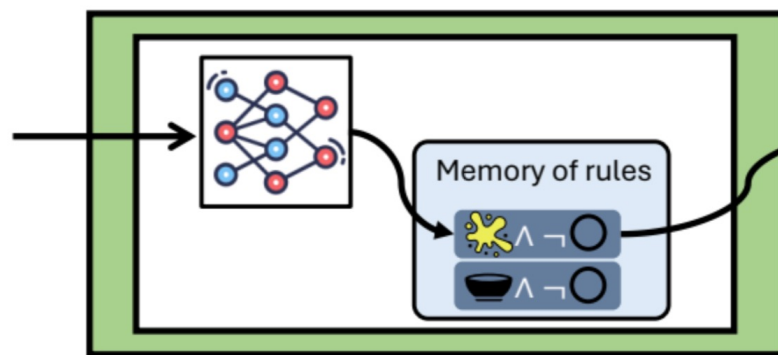
where the mixing coeff. are parametrized by the selector



CONCEPT-BASED MEMORY REASONING

CMR has 3 key features:

- **Universal approximator** akin to opaque DNNs (Theorem 4.1)
- Minimize #relevant concepts $\rightarrow$ understanding is **tractable**



Inference mechanisms can only be **selected** from a finite set of **transparent rules**!

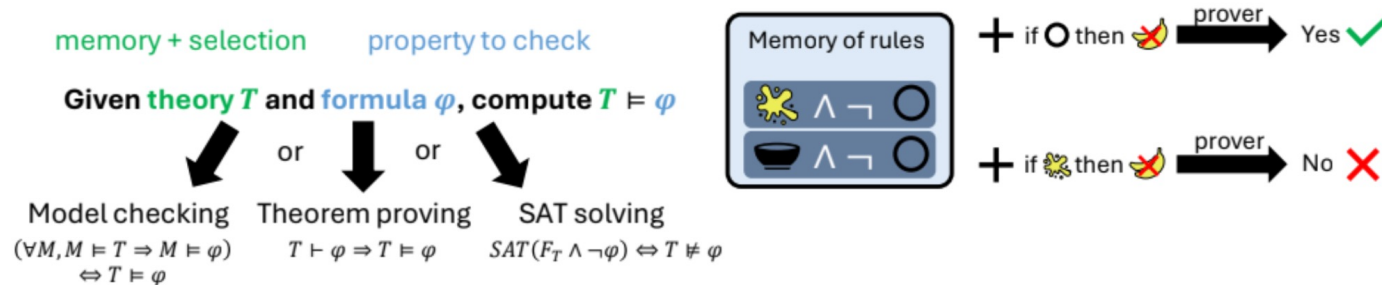


CONCEPT-BASED MEMORY REASONING

CMR has 3 key features:

- **Universal approximator** akin to opaque DNNs (Theorem 4.1)
- Minimize #relevant concepts $\rightarrow$ understanding is **tractable**
- The concept memory allows **formal verification** of properties

"Does a property hold no matter which rule is selected?"



(INFORMAL) OBJECTIVE

Build a general-purpose neural model that is:

✓ **Expressive** (as DNNs)

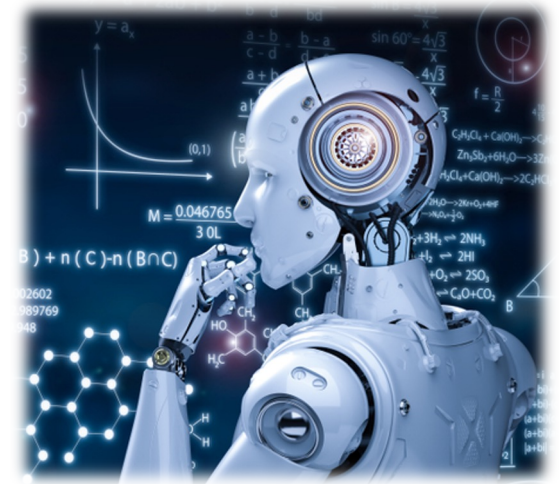
and whose inference mechanism is

✓ **Functionally transparent** (we fully understand CPT/TT)

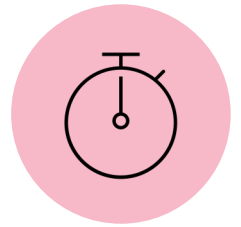
✓ **Tractable** (CPT/TT size $\ll$ exp)

✓ **Semantically transparent** (concept-based)

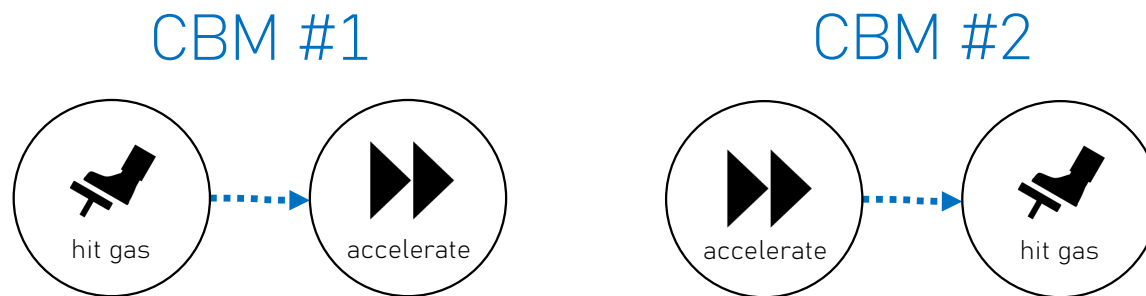
❑ **Causally transparent** (based on non-trivial cause-effect chains)



COMPARING BOTTLENECKS



Is CBM #1 more causally transparent than CBM #2?



If yes, raise your hand!

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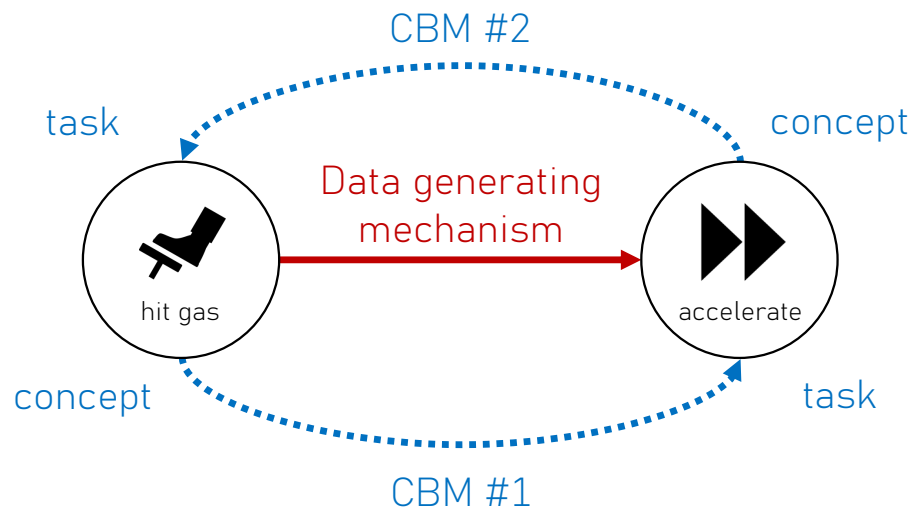
CAUSAL OPACITY

- **Causal reliability**: discover causal mechanisms of the **data generating process**



CAUSAL OPACITY

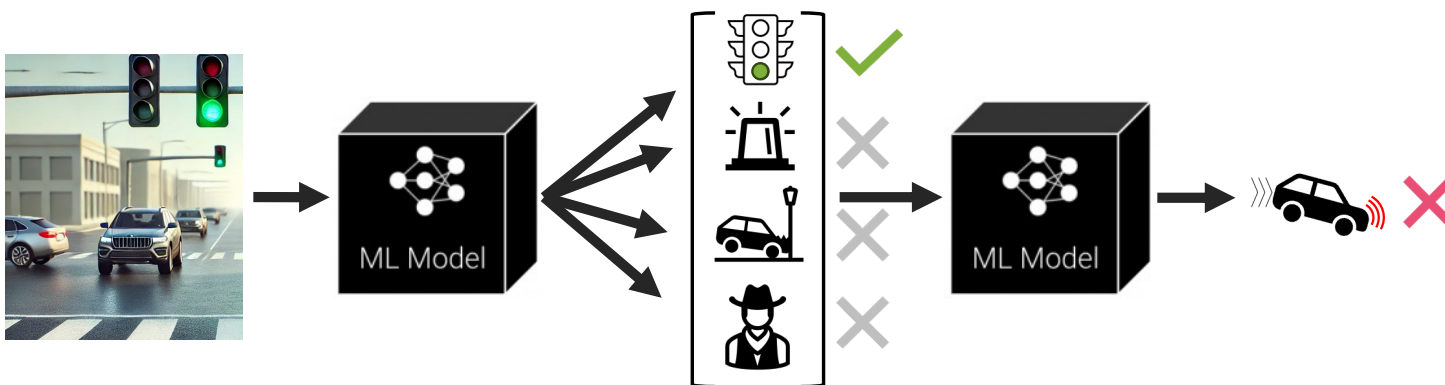
- **Causal reliability**: discover causal mechanisms of the **data generating process**
- **Causal opacity**: discover causal mechanism of a **model's inference process**



CBM #1 is potentially equally transparent w.r.t. CBM #2!

MEASURING CAUSAL EFFECTS

CBMs can **answer association** queries (duh...)



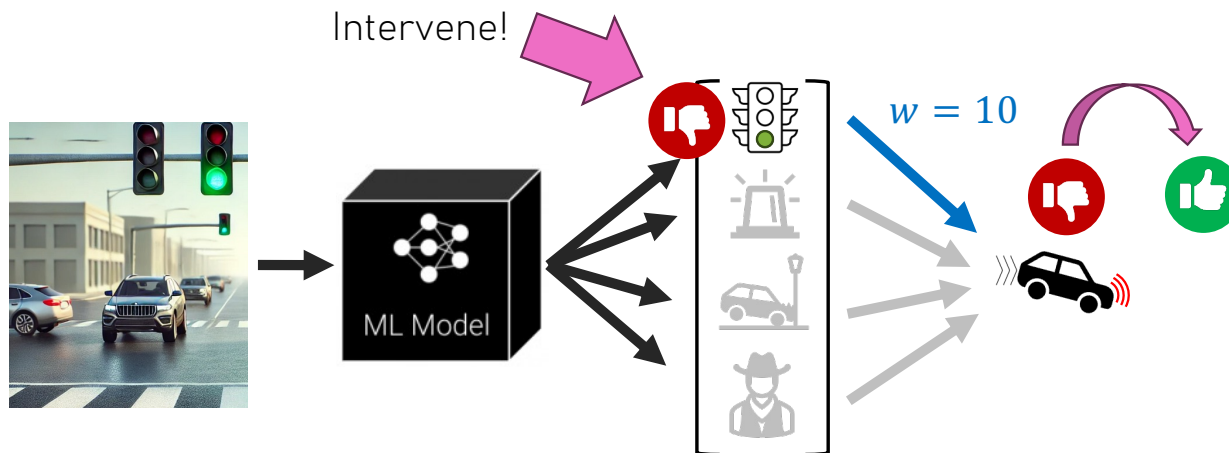
Association

What if the model sees a green light?

$P(\text{brake} \mid \text{light})$

MEASURING CAUSAL EFFECTS

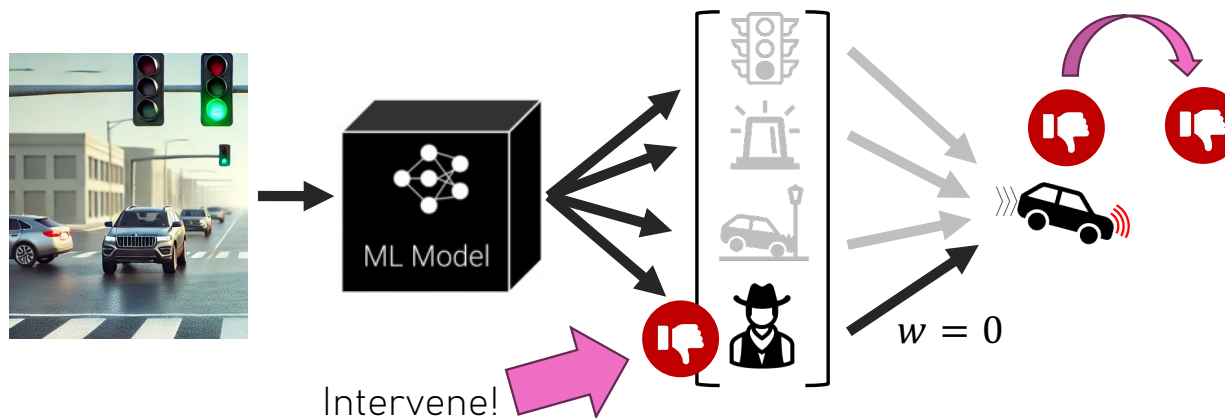
Sometimes intervening on wrongly predicted concepts helps...



MEASURING CAUSAL EFFECTS

Sometimes intervening on wrongly predicted concepts helps...
and sometimes it doesn't! 😞

Causal analysis can provide us with insights!



Yes, but...

MEASURING CAUSAL EFFECTS

Sometimes intervening on wrongly predicted concepts helps...
and sometimes it doesn't! 😞

*Can we measure the causal influence of a
concept on the task?*



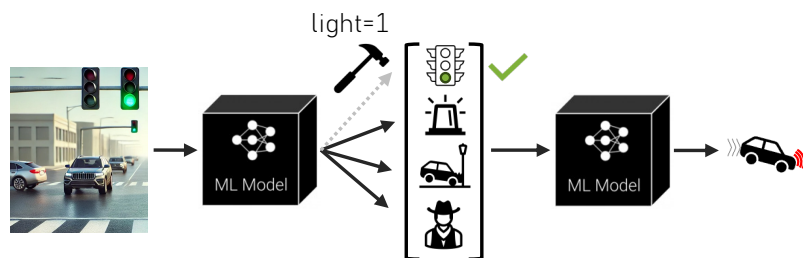
Yes, but...

CAUSAL CONCEPT EFFECT



Proposed Solution

Step 1: Compute **expected value** of the task with $do(c_i = 1)$



$$\mathbb{E}[\text{brake} \mid do(\text{light} = 1)] = 0.2$$

Intervention

What if I set the light color to red?
 $P(\text{brake} \mid do(\text{light}))$

Association

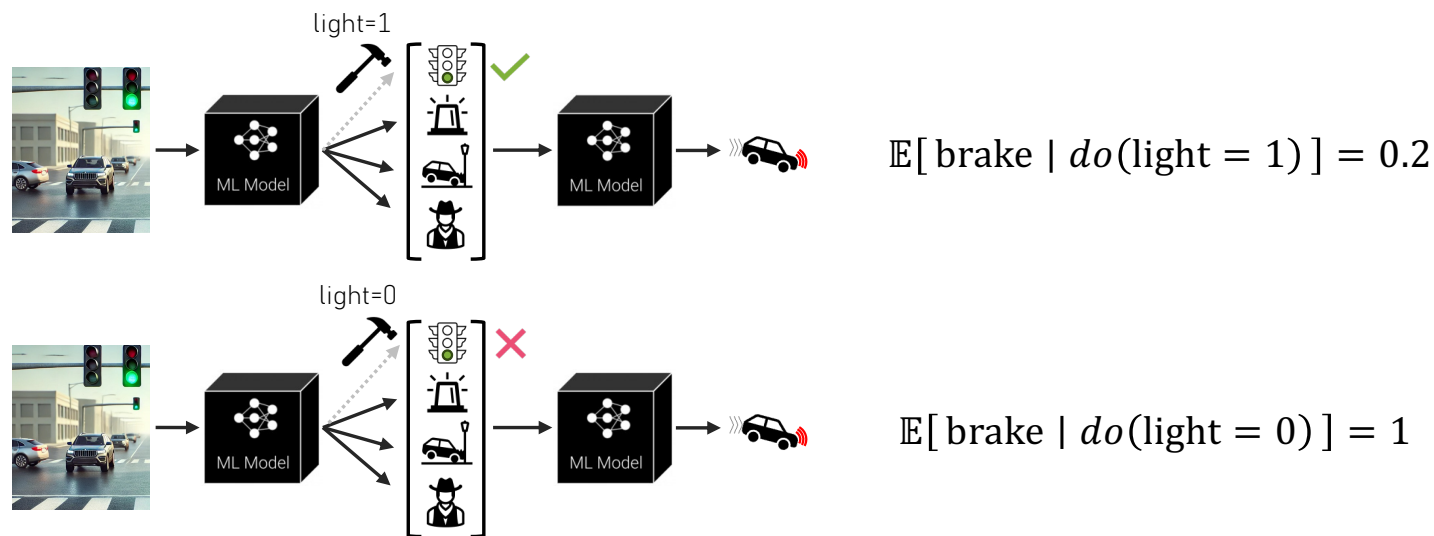
What if the model sees a green light?
 $P(\text{brake} \mid \text{light})$

CAUSAL CONCEPT EFFECT



Proposed Solution

Step 2: Compute **expected value** of the task with $do(c_i = 0)$



Intervention

What if I set the light color to red?
 $P(\text{brake} \mid do(\text{light}))$

Association

What if the model sees a green light?
 $P(\text{brake} \mid \text{light})$

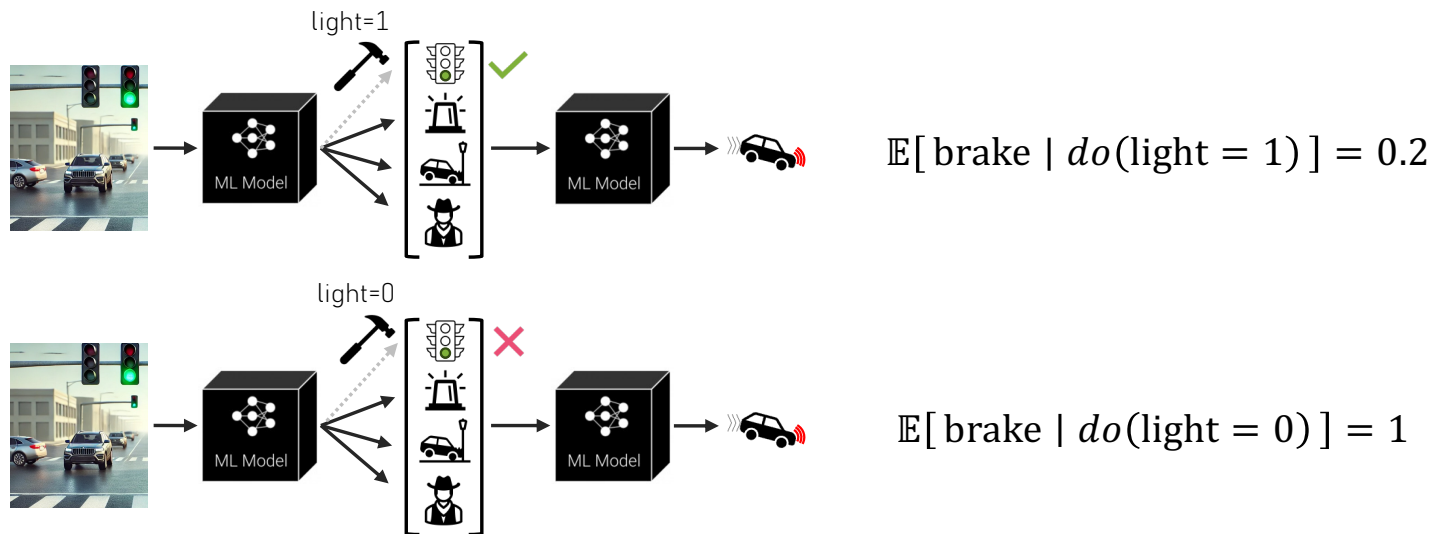
CAUSAL CONCEPT EFFECT



Proposed Solution

Step 3: Compute **difference of expected values**: absolute value is proportional to **causal effect**

$$\text{CaCE} = \mathbb{E}[\text{brake} \mid \text{do}(\text{light} = 1)] - \mathbb{E}[\text{brake} \mid \text{do}(\text{light} = 0)] = -0.8$$



Intervention

What if I set the light color to red?
 $P(\text{brake} \mid \text{do}(\text{light}))$

Association

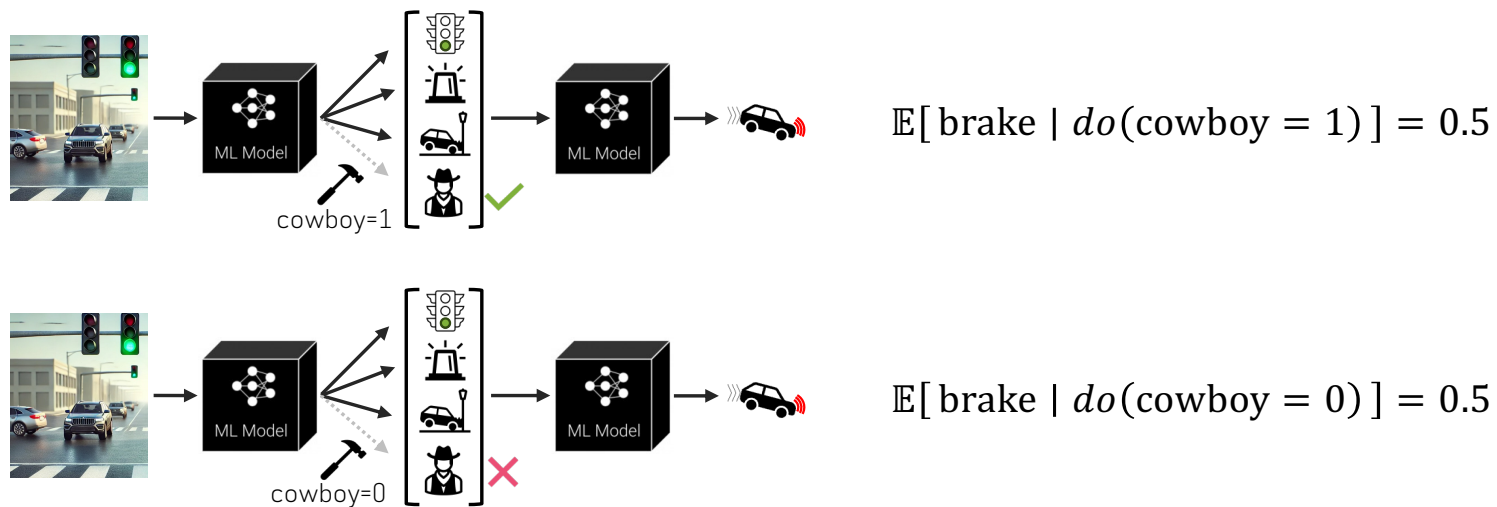
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 $P(\text{brake} \mid \text{light})$

CAUSAL CONCEPT EFFECT



Proposed Solution

Step 3: Compute **difference of expected values**: absolute value is proportional to **causal effect**



Intervention

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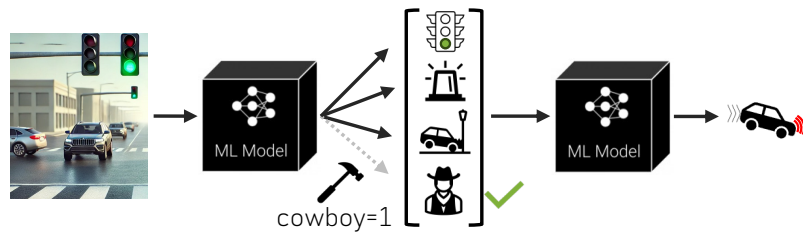
CAUSAL CONCEPT EFFECT



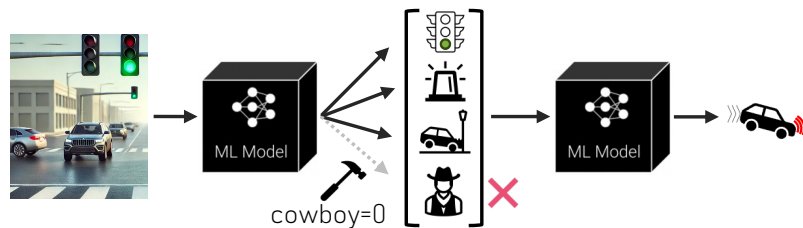
Proposed Solution

Step 3: Compute **difference of expected values**: absolute value is proportional to **causal effect**

$$\text{CaCE} = \mathbb{E}[\text{brake} \mid \text{do}(\text{cowboy} = 1)] - \mathbb{E}[\text{brake} \mid \text{do}(\text{cowboy} = 0)] = 0$$



$$\mathbb{E}[\text{brake} \mid \text{do}(\text{cowboy} = 1)] = 0.5$$



$$\mathbb{E}[\text{brake} \mid \text{do}(\text{cowboy} = 0)] = 0.5$$

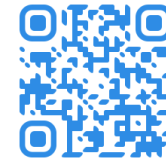
Intervention

What if I set the light color to red?
 $P(\text{brake} \mid \text{do}(\text{light}))$

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What if the model sees a green light?
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COUNTERFACTUAL CBMS



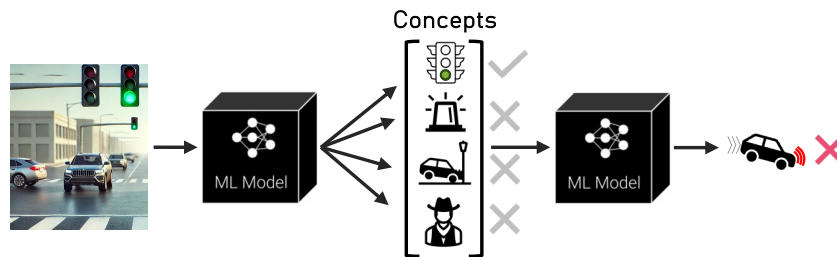
ICLR25



ICML22

Limitation Being Addressed

CBMs cannot answer **counterfactual queries**!



Counterfactual

What would have been predicted in the same circumstance had a car crash be seen?

$$P(\text{brake} \mid \text{light}, \text{crash})$$

Intervention

What if I set the light color to red?

$$P(\text{brake} \mid \text{do}(\text{light}))$$

Association

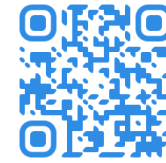
What if the model sees a green light?

$$P(\text{brake} \mid \text{light})$$

[1] Dominici, Gabriele, et al. "Counterfactual Concept Bottleneck Models." ICLR 2025.

[2] Abid, Abubakar, et al. "Meaningfully debugging model mistakes using conceptual counterfactual explanations." International Conference on Machine Learning. PMLR, 2022.

COUNTERFACTUAL CBMS



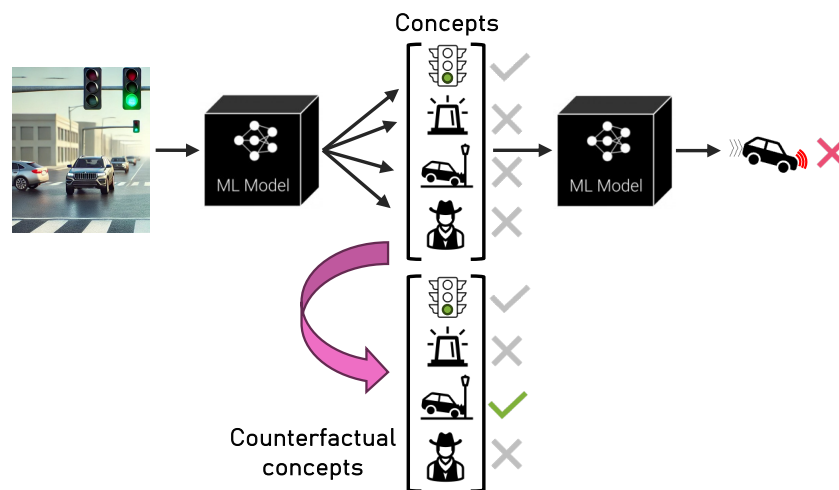
ICLR25



ICML22

Proposed Solution

Step 1: Generate counterfactual concept activations



Counterfactual

What would have been predicted in the same circumstance had a car crash be seen?

$$P(\text{brake} \mid \text{light}, \text{crash})$$

Intervention

What if I set the light color to red?

$$P(\text{brake} \mid \text{do}(\text{light}))$$

Association

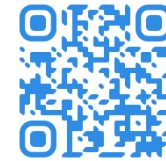
What if the model sees a green light?

$$P(\text{brake} \mid \text{light})$$

[1] Dominici, Gabriele, et al. "Counterfactual Concept Bottleneck Models." ICLR 2025.

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COUNTERFACTUAL CBMS



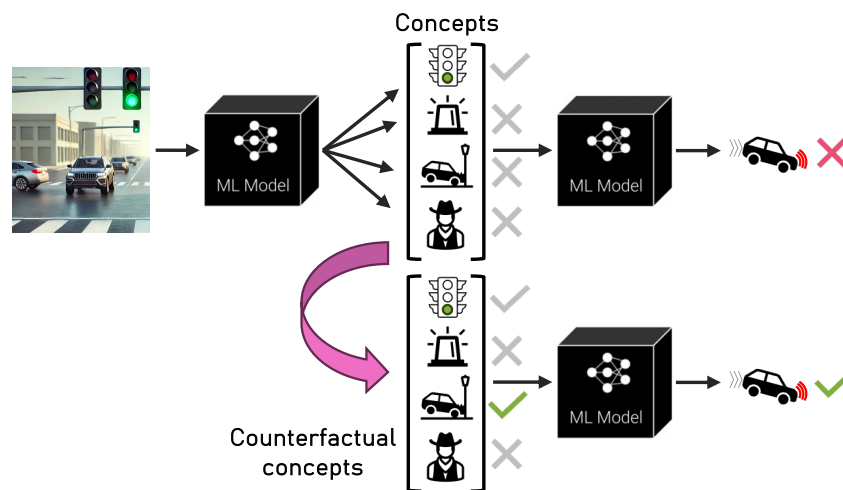
ICLR25



ICML22

Proposed Solution

Step 2: Compute **causal effect** on the task!



Counterfactual

What would have been predicted in the same circumstance had a car crash be seen?

$$P(\text{brake} \mid \text{light}, \text{crash})$$

Intervention

What if I set the light color to red?

$$P(\text{brake} \mid \text{do}(\text{light}))$$

Association

What if the model sees a green light?

$$P(\text{brake} \mid \text{light})$$

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DIRECT COUNTERFACTUAL DEPENDENCE

So far, we have been making 2 **strong assumptions**...

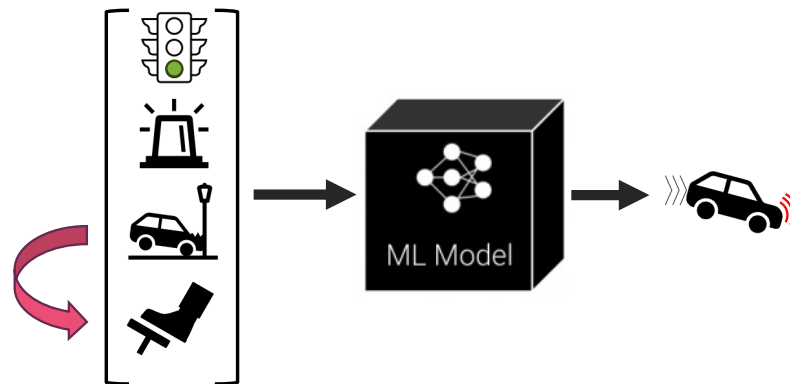


DIRECT COUNTERFACTUAL DEPENDENCE

So far, we have been making 2 **strong assumptions**:

- Concepts are **mutually independent**

Intervening on "car crash" does not increase the likelihood of hitting the brakes!



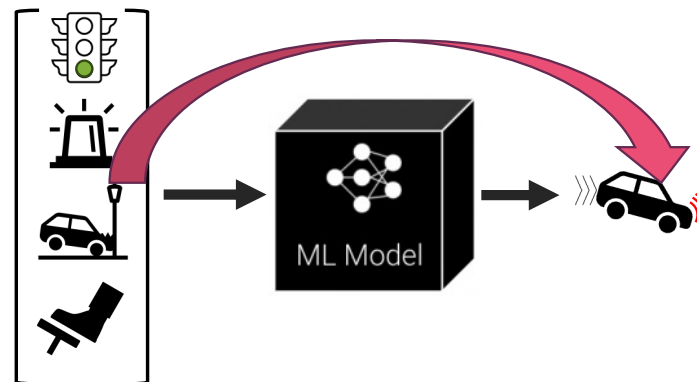
DIRECT COUNTERFACTUAL DEPENDENCE

So far, we have been making 2 **strong assumptions**:

- Concepts are **mutually independent**
- Concepts are **direct causes** of the task

Intervening on "car crash" does not increase the likelihood of hitting the brakes!

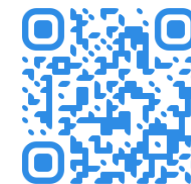
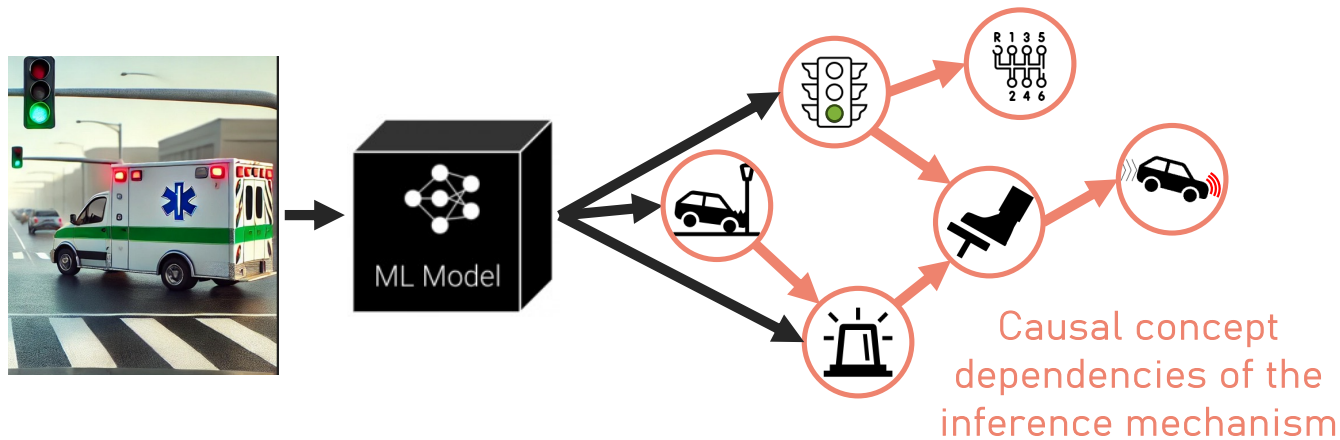
Intervening on "car crash" directly causes the car to brake!



CONCEPT GRAPH MODELS

Proposed Solution

Enforce inference through a **concept graph**!



ICLR25



CLearR24

[1] Dominici, Gabriele et al. "Causal Concept Graph Models: Beyond Causal Opacity in Deep Learning." ICLR 2025.

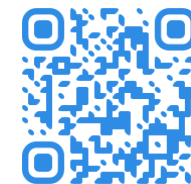
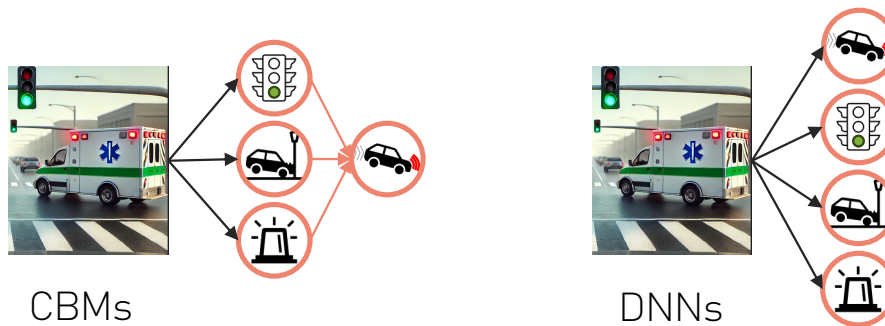
[2] Moreira, Ricardo et al. "Diconstruct: Causal concept-based explanations through black-box distillation." CLearR 2024.

CONCEPT GRAPH MODELS

Proposed Solution

The concept graph can be:

- Given as a prior



ICLR25



CLearR24

[1] Dominici, Gabriele et al. "Causal Concept Graph Models: Beyond Causal Opacity in Deep Learning." ICLR 2025.

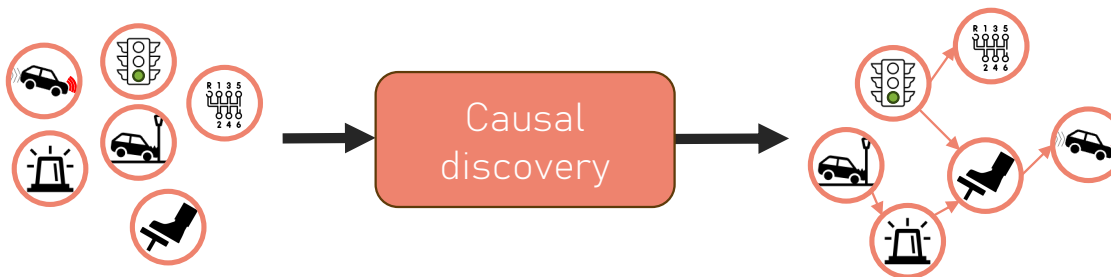
[2] Moreira, Ricardo et al. "Diconstruct: Causal concept-based explanations through black-box distillation." CLearR 2024.

CONCEPT GRAPH MODELS

Proposed Solution

The concept graph can be:

- Given as a prior
- Extracted from data with causal discovery techniques



ICLR25



CLear24

[1] Dominici, Gabriele et al. "Causal Concept Graph Models: Beyond Causal Opacity in Deep Learning." ICLR 2025.

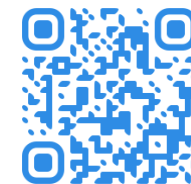
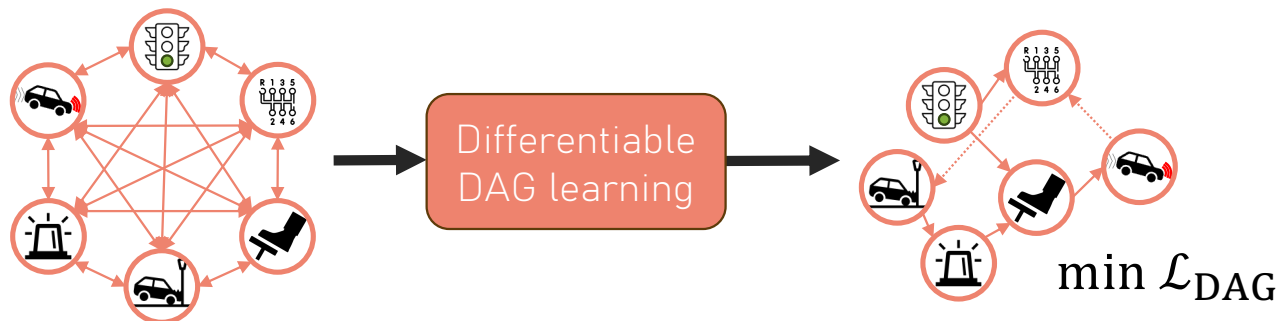
[2] Moreira, Ricardo et al. "Diconstruct: Causal concept-based explanations through black-box distillation." CLear 2024.

CONCEPT GRAPH MODELS

Proposed Solution

The concept graph can be:

- Given as a prior
- Extracted from data with causal discovery techniques
- Obtained with differentiable DAG learning



ICLR25



CLear24

[1] Dominici, Gabriele et al. "Causal Concept Graph Models: Beyond Causal Opacity in Deep Learning." ICLR 2025.

[2] Moreira, Ricardo et al. "Diconstruct: Causal concept-based explanations through black-box distillation." CLear 2024.

NEURAL INTERPRETABLE REASONING

Build a **general-purpose** neural model that is:

✓ **Expressive** (as DNNs)

and whose inference mechanism is

✓ **Semantically transparent** (concept-based)

✓ **Functionally transparent** (we fully understand CPT/TT)

✓ **Tractable** (CPT/TT size $\ll$ exp)

✓ **Causally transparent** (based on non-trivial cause-effect chains)

